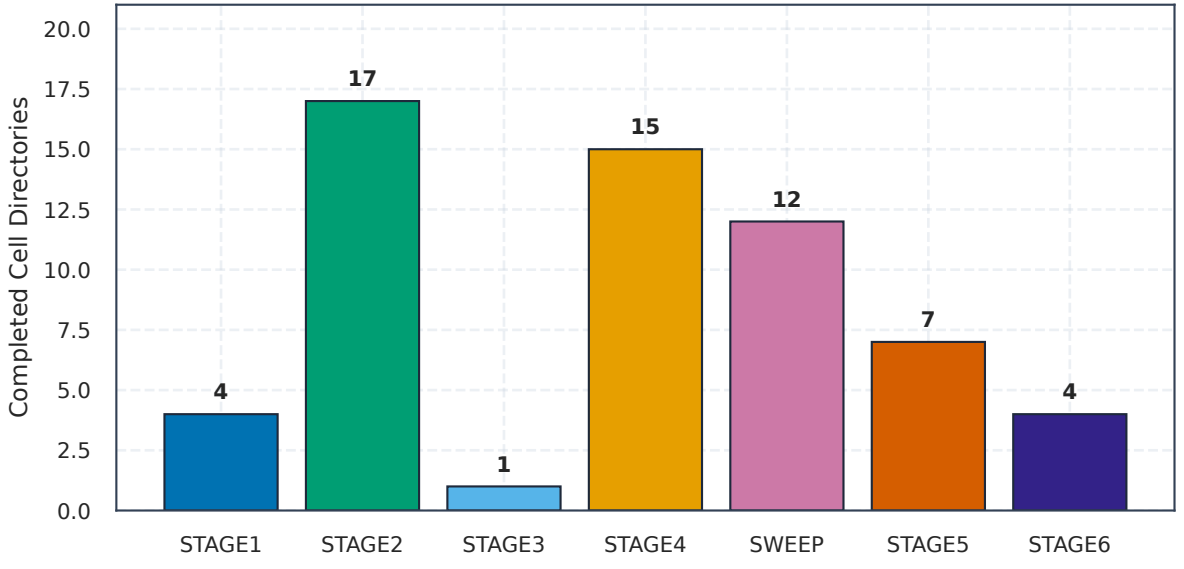
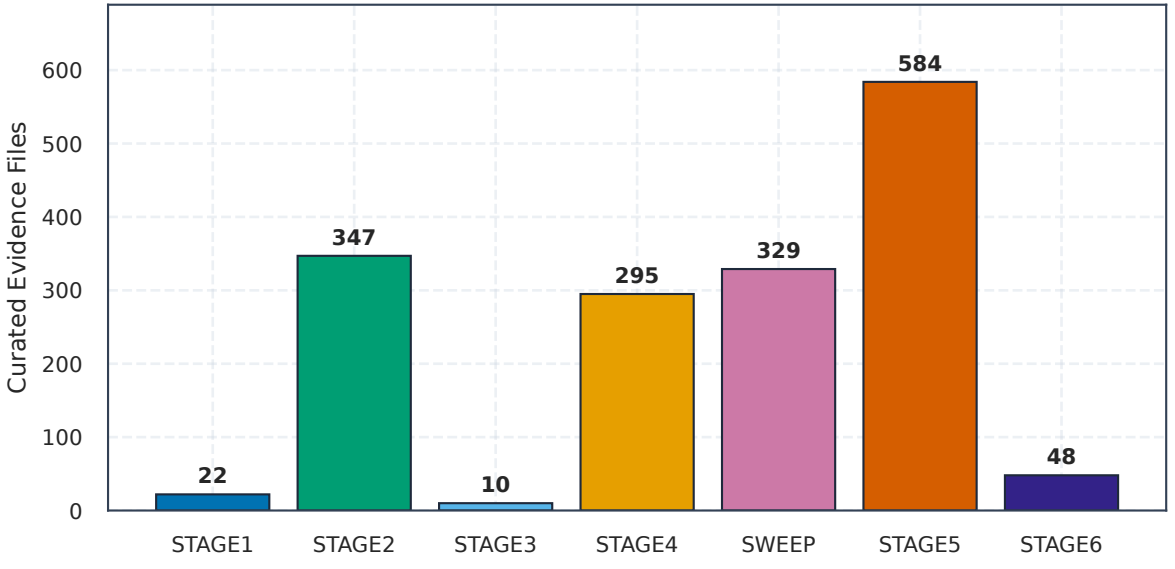
A. Six-Stage, 60-Directory GCP Empirical Study Architecture (VarDictRust v1.0.6, FP=5865a2a0b24a8c17)

Stage 1: Ceiling & Tiling	Stage 2: Tier P Panels	Stage 3: Tier G RNA-Seq	Stage 4: Scaling & ROC	Option C: Parity Sweep	Stage 5: Full WGS & 10-Chem	Stage 6: Next-Gen Truth
(H1a 0.5-10M, Gate 7)	(Wolff 2026, -R chr1, 13m)	(51.8 GiB HG002 mRNA -i)	(1-32c Exome, chr1, QUAL)	(.var/.vcf & MRD -f 0)	(HG002/008 Arm A/B, H4)	(v5.0Q, CMRG, Mosaic, v0.4)
4 dirs | 22 files **100% rc=0 PASS**	**17 dirs | 347 files** **100% rc=0 PASS**	**1 dirs | 10 files** **100% rc=0 PASS**	**15 dirs | 295 files** **100% rc=0 PASS**	**12 dirs | 329 files** **100% rc=0 PASS**	**7 dirs | 584 files** **100% rc=0 PASS**	**4 dirs | 48 files** **100% rc=0 PASS**

Hardware Lock: GCP n2-standard-32 (Intel Xeon Platinum 8373C Ice Lake, 32 vCPUs, 128 GiB RAM, 4x375 GB NVMe RAID0) | Comparators: VarDictJava v1.8.3 (0886261) & vardictcpp ae93fe39dc (Wolff 2026)
Cloud Spend: 228.90/**300.00 Budget (760.96 GiB GCS Cache; 60/60 Cells Complete)**

B. Completed GCP Directories per Stage (Total = 60)



C. Verified Output Artifacts per Stage (Time, Verdict & Truth Files)